HW 0324

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1.
(a)

$$
\begin{aligned}
SST &= \sum (y_i - \bar{y})^2 \\
&= \sum y_i^2 - N\bar{y}^2 \\
&= 7825.94 - 20 \times 369.0241 \\
&= 7825.94 - 7380.482 \\
&= 445.458
\end{aligned}
$$

$$
R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} \approx 0.8429
$$

(b)

$$
\begin{aligned}
\hat{\sigma}^2 &= \frac{SSE}{N-2} \\
&= \frac{SST \times (1 - R^2)}{N-2} \\
&= \frac{725.94 \times (1 - 0.7911)}{20 - 2} \\
&= \frac{151.648866}{18} \approx 8.4249
\end{aligned}
$$

(c)

$$
\begin{aligned}
R^2 &= 1 - \frac{SSE}{SST} \\
&= 1 - \frac{\sum \hat{e}_i^2}{\sum (y_i - \bar{y})^2} \\
&= 1 - \frac{182.85}{631.63} \\
&= 1 - 0.289489 \approx 0.7105
\end{aligned}
$$

3.

(a)

$$
\hat{y}_0 = 1.2 + 0.8(4) = 1.2 + 3.2 = 4.4
$$

(b)

$$
\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{5-2} = 1.2
$$

$$\hat{\sigma} = \sqrt{1.2} \approx 1.0954, \quad \bar{x} = \frac{3+2+1-1+0}{5} = \frac{5}{5} = 1$$

$$\begin{aligned} se(f) &= \hat{\sigma}\sqrt{1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum(x_i - \bar{x})^2}} \\ &= 1.0954 \times \sqrt{1 + \frac{1}{5} + \frac{(4-1)^2}{10}} \\ &= 1.5874 \end{aligned}$$

(c)

$$t_{c,(N-2)} = t_{0.025,3} = 3.182$$

$$\begin{aligned} 95\% \text{ prediction interval} &= [\hat{y}_0 - t_{c,(N-2)} \times se(f), \hat{y}_0 + t_{c,(N-2)} \times se(f)] \\ &= [4.4 - 3.182 \times 1.5874, 4.4 + 3.182 \times 1.5874] \\ &= [-0.651, 9.451] \end{aligned}$$

(d)

$$t_{c,(N-2)} = t_{0.005,3} = 5.841$$

$$\begin{aligned} 99\% \text{ prediction interval} &= [\hat{y}_0 - t_{c,(N-2)} \times se(f), \hat{y}_0 + t_{c,(N-2)} \times se(f)] \\ &= [4.4 - 5.841 \times 1.5874, 4.4 + 5.841 \times 1.5874] \\ &= [-4.872, 13.672] \end{aligned}$$

(e)

$$x_0 = \bar{x} = 1, \hat{y} = 1.2 + 0.8(1) = 2.0$$

$$se(f)_{\bar{x}} = 1.0954 \times \sqrt{1 + \frac{1}{5} + 0} = 1.0954 \times 1.0954 = 1.2$$

$$t_{c,(N-2)} = t_{0.025,3} = 3.182$$

$$\begin{aligned} 95\% \text{ prediction interval} &= [\hat{y}_0 - t_{c,(N-2)} \times se(f), \hat{y}_0 + t_{c,(N-2)} \times se(f)] \\ &= [2.0 - 3.182 \times 1.2, 2.0 + 3.182 \times 1.2] \\ &= [-1.8184, 5.8184] \end{aligned}$$

The interval in (e) is smaller than that in (c).

25.

(a)

The estimated intercept ($\hat{\beta}_1$) is 4.39, which represents the predicted value of Y when X is zero. The price is $e^{4.39} = 80.6404$
The slope coefficient ($\hat{\beta}_2$) is 0.036, suggesting that for every one-unit increase in SQFT, the PRICE increases 3.6% on average.

$$\frac{d\ln(PRICE)}{dSQFT} = \beta_2 \implies \frac{1}{PRICE} \cdot \frac{dPRICE}{dSQFT} = \beta_2$$

$$Slope = \frac{dPRICE}{dSQFT} = \beta_2 \times \overline{PRICE} = 0.036 * 250.2369 = 9.0085$$

Table 1:

	Dependent variable:
	log(price)
sqft	0.036***
	(0.001)
Constant	4.394***
	(0.043)
Observations	500
R^2	0.542
Adjusted R^2	0.541
Residual Std. Error	0.340 (df = 498)
F Statistic	588.686*** (df = 1; 498)
Note:	*p<0.1; **p<0.05; ***p<0.01

$$\epsilon = \frac{dPRICE}{dSQFT} \cdot \frac{SQFT}{PRICE} = \beta_2 \times \overline{SQFT} = 0.9822$$

(b)

Table 2:

	Dependent variable:
	log(price)
log(sqft)	1.025***
	(0.048)
Constant	2.050***
	(0.158)
Observations	500
R^2	0.474
Adjusted R^2	0.473
Residual Std. Error	0.364 (df = 498)
F Statistic	448.488*** (df = 1; 498)
Note:	*p<0.1; **p<0.05; ***p<0.01

The estimated intercept ($\hat{\beta}_1$) is 2.05, which represents the price is $e^{2.05} = 7.7679$ when $SQFT = 1(\ln(SQFT) = 0)$.
The slope coefficient ($\hat{\beta}_2$) is 1.025, suggesting that for SQFT increases 1% , the PRICE increases 1.025% on average.

$$\frac{dPRICE}{dSQFT} = \frac{dPRICE}{d\ln(PRICE)} \cdot \frac{d\ln(PRICE)}{d\ln(SQFT)} \cdot \frac{d\ln(SQFT)}{dSQFT}$$

$$Slope = \frac{dPRICE}{dSQFT} = PRICE \cdot \beta_2 \cdot \frac{1}{SQFT} = \beta_2 \left(\frac{\overline{PRICE}}{\overline{SQFT}} \right) = 9.4015$$

$$\epsilon = \frac{d\ln(PRICE)}{d\ln(SQFT)} = \beta_2 = 1.025$$

(c)
R-square in linear model: 0.6413167
R-square in (b): 0.6621612
R-square in (c): 0.6445084

Among the three specifications, the Log-Linear model (b) yields the highest Generalized R^2 (0.6622), suggesting that it provides the best fit for the house price data in Baton Rouge. This improvement over the linear model (0.6413) indicates that the relationship between house prices and square footage is better captured through a semi-elasticity framework, likely due to the mitigation of heteroskedasticity inherent in real estate data.

(d)

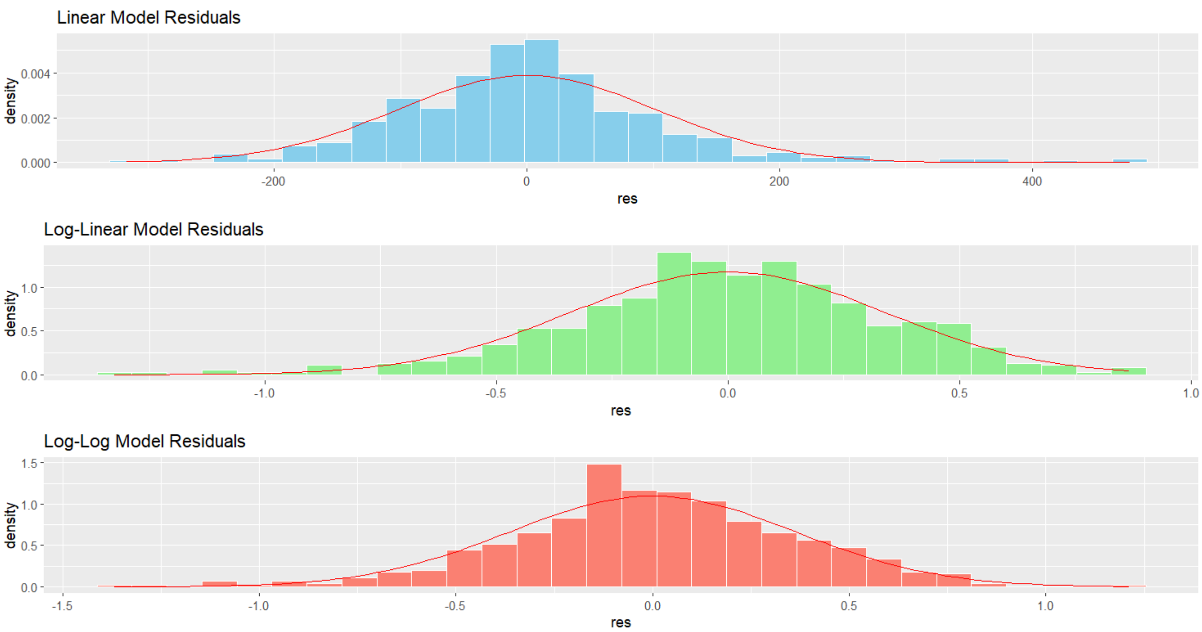
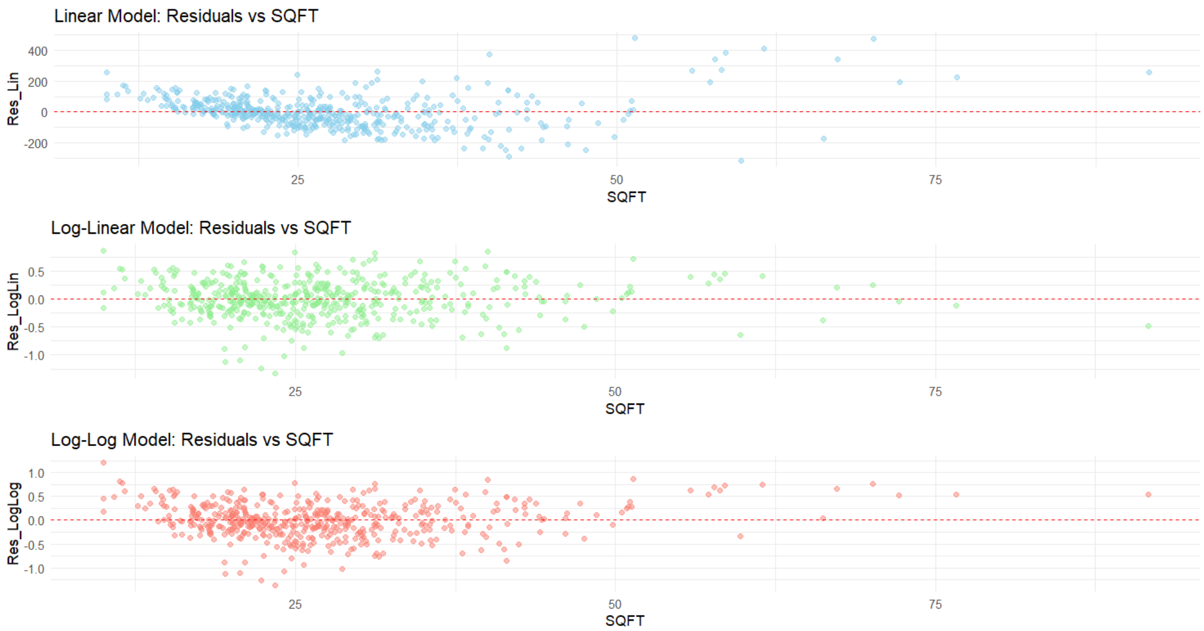


Table 3: Jarque-Bera Test Results for Residual Normality

Model	JB Statistic (χ^2)	p-value	Normality (H_0)
(a) Linear	221.11	$<$ 2.2e-16	Rejected
(b) Log-Linear	26.68	1.61e-06	Rejected
(c) Log-Log	14.28	0.00079	Rejected

The p-values for all three models are less than 0.05, leading to the rejection of the null hypothesis of normality. The distribution of the residuals is not compatible with the assumption of normality.

(e)



Residual analysis indicates that none of the specifications fully eliminate heteroskedasticity. While the linear model suffers from variance that scales with house size, the log-linear and log-log models exhibit an overcorrection effect, where residuals for smaller homes show disproportionately higher variance compared to larger ones.

(f)

Table 4: Predicted Prices and 95% Prediction Intervals (Units: $1,000)

Model Specification	Predicted Price ($\hat{y}$)	95% Lower PI	95% Upper PI
(a) Linear Model	246.46	44.28	448.63
(b) Log-Linear Model	214.23	109.77	418.12
(c) Log-Log Model	227.54	111.14	465.84

(a) 214.2336 , (b) 227.5386, (c) 246.4557 (thousands of dollars)

(g) (a) [109.7685, 418.1167] , (b) [111.1406, 465.8406], (c)[44.27727, 448.6341] (thousands of dollars)

(h)

Choice of Functional Form:

Despite the rejection of normality by the Jarque-Bera tests across all specifications, the **Log-Linear model** is deemed the most suitable for the following reasons:

1. It exhibits the highest *Generalized* R^2, demonstrating superior predictive accuracy.
2. It significantly compresses the variance stemming from price outliers, addressing the heteroskedasticity issues inherent in the linear form.
3. It satisfies the non-negativity constraint, ensuring that all point estimates and prediction intervals are strictly positive ($\hat{y} > 0$).